

Topological Quantum Matters (TQM)

Time:

Every Mon. & Wed. 13:30-15:05, from 2021-9-13 to 12-3 (weeks 1-12 of the fall semester in Tsinghua)

Venue:

It will be a combination of offline (宁斋W11, Ning Zhai W11) and online (腾讯会议tencent meeting: 5772849861, password: 654321)

Description:

In this course, we will use topology to understand some exotic quantum phases of matter. The topics will include topological insulators, topological orders, symmetry-protected topological phases, etc. The course will cover both condensed matter models in physics and general mathematical descriptions (such as group cohomology theory and modular tensor categories of knots).

Prerequisites:

Basic topology and quantum mechanics. We will try to make a compromise between mathematics and physics by introducing relevant concepts self-consistently, as there are audience from both sides.

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Other Courses :

Peshtikhim : Inv. of knots and 3-mfd.

Hashigata : Topological Quantum Computation.

Zheng : Category theory

Introduction to QM.

c) Why important to real life?

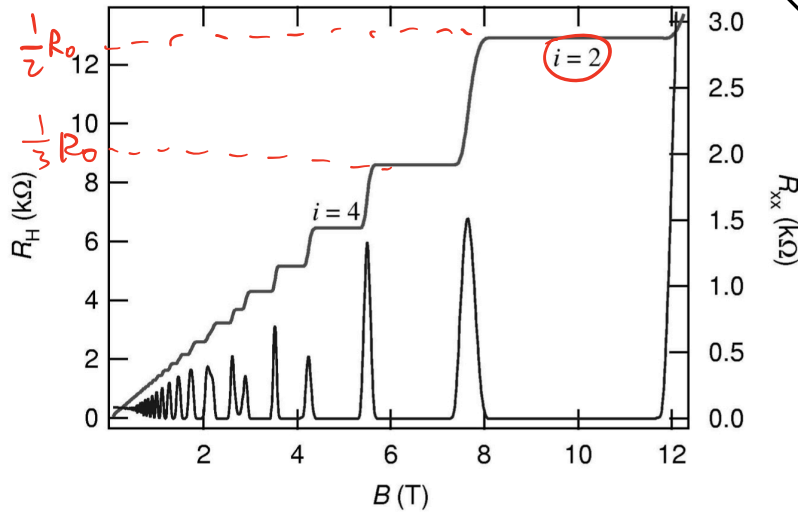
One example:

Q: how to measure { electric charge e
Planck constant h accurately? }

① Integer
Quantum Hall effect

Nobel 1985

$$R_0 = \frac{h}{e^2}$$



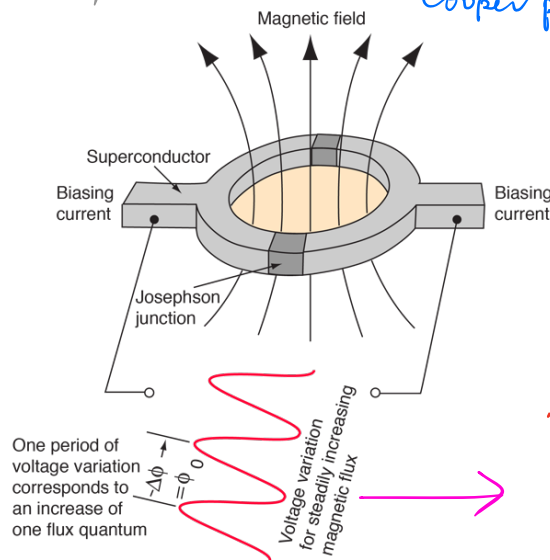
$$\begin{cases} e = \frac{2 \Phi_0}{R_0} \\ h = \frac{4 \Phi_0^2}{R_0} \end{cases}$$

② Josephson effect

Nobel 1973

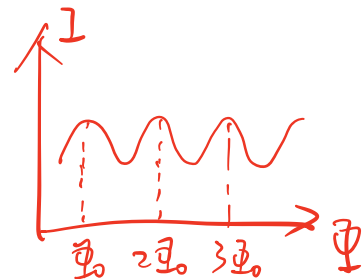
$$\Phi_0 = \frac{h}{2e}$$

Cooper pair



International
System of
Units
(before 2018)

[After 2018, e, h are fixed by definition]



using many-body system to measure properties of a single electron!
accurately

(2) Why exotic?

fundamental charge e = charge of one single electron/proton/...

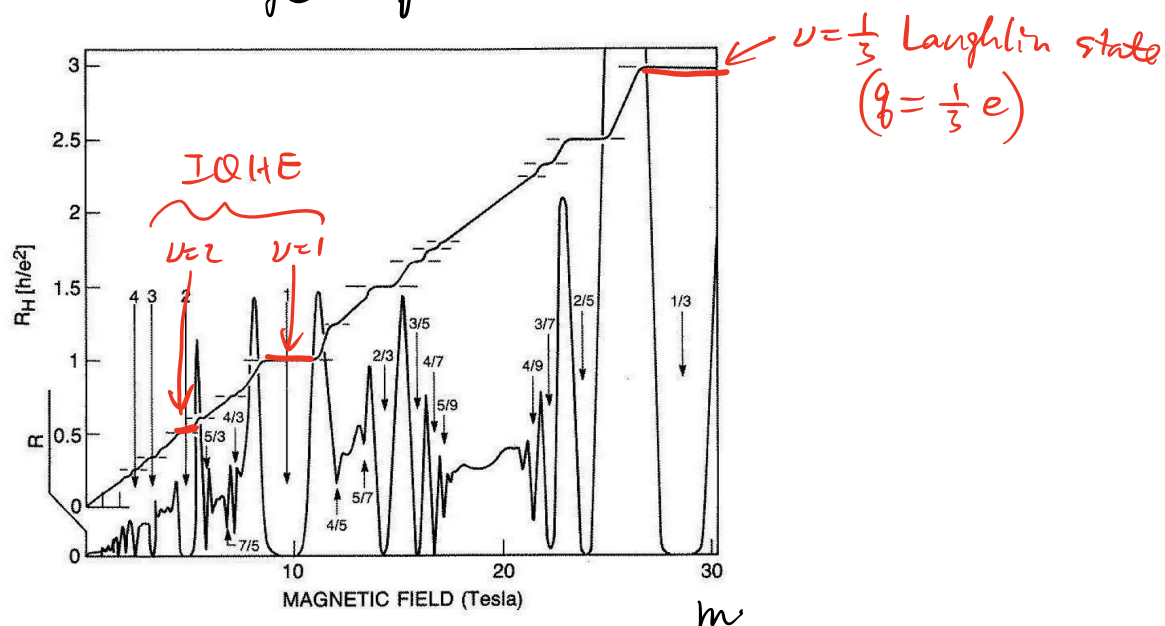
Fractional quantum Hall effect (FQHE) Nobel 1998



$$R = \frac{h}{\nu e^2}$$

$$\begin{cases} \nu \in \mathbb{Z}: & \text{integer QHE} \\ \nu \in \mathbb{Q}: & \text{FQHE} \end{cases}$$

Fractional charge $q = \nu \cdot e$



Fractional charge appears in many-body systems!

(3) Why interesting to many people?

related to many other topics:

- math: knot theory, Jones polynomial, Turaev-Viro inv., Reshetikhin-Turaev inv., modular tensor categories, 2-knot, higher categories ...
lecture Zheng's lecture
- physics: IQHE, FQHE, topological insulator, ...
(real materials)
- quantum information: topological quantum computation
(by Freedman, Kitaev)
fault-tolerant quantum computation

(4) Why hard/deep?

$\propto$ number of degrees of freedom

$\propto$ dimensional space.

foundations of quantum field theory (= many-body quantum mechanics)

emergence of spacetime/gravity (?)

Syllabus (tentative):

It may vary depending on the actual speed of the course. By weeks (4*45min/week, 12 weeks):

- (1) introduction to TQM, different classes of TQM (bosonic/fermionic, long/short range entangled, with/without symmetry),
1st example: Kitaev's toric code model (homology enters),
2nd example: Haldane's honeycomb model (homotopy enters) *1988* $\rightsquigarrow$ *Qikun Xue : quantum anomalous Hall effect. 2013*

Part I. Bosonic topological orders

- (2) quantum double model, twisted quantum double model = Dijkgraaf-Witten gauge theory
- (3) introduction to fusion categories, Levin-Wen model = Turaev-Viro model
- (4) introduction to modular tensor categories, general description of anyon models by Kitaev
- (5) 3+1D Walker-Wang model = Crane-Yetter model

Part II. Topological insulators (fermionic symmetry-protected topological phases without interactions)

- (6) introduction to band theory, integer quantum Hall effect, Thouless-Kohmoto-Nightingale-den Nijs number, Chern insulator
TKNN : Hall conductance = Chern number
- (7) examples: Kitaev's Majorana chain, Su-Schrieffer-Heeger model, 2+1D and 3+1D topological insulators, edge theories
- (8) symmetries in free fermion system, 10-fold way classification $\rightarrow$ *topological K theory, Clifford algebra*

Part III. Symmetry-protected topological phases

- (9) introduction to symmetry-protected topological (SPT) phases, Haldane chain
- (10) introduction to projective representation, tensor product state, classification of 1+1D bosonic SPT
- (11) Levin-Gu model, introduction to group cohomology, bosonic SPT model from group cohomology
- (12) introduction to fermionic SPT phases, other related topics

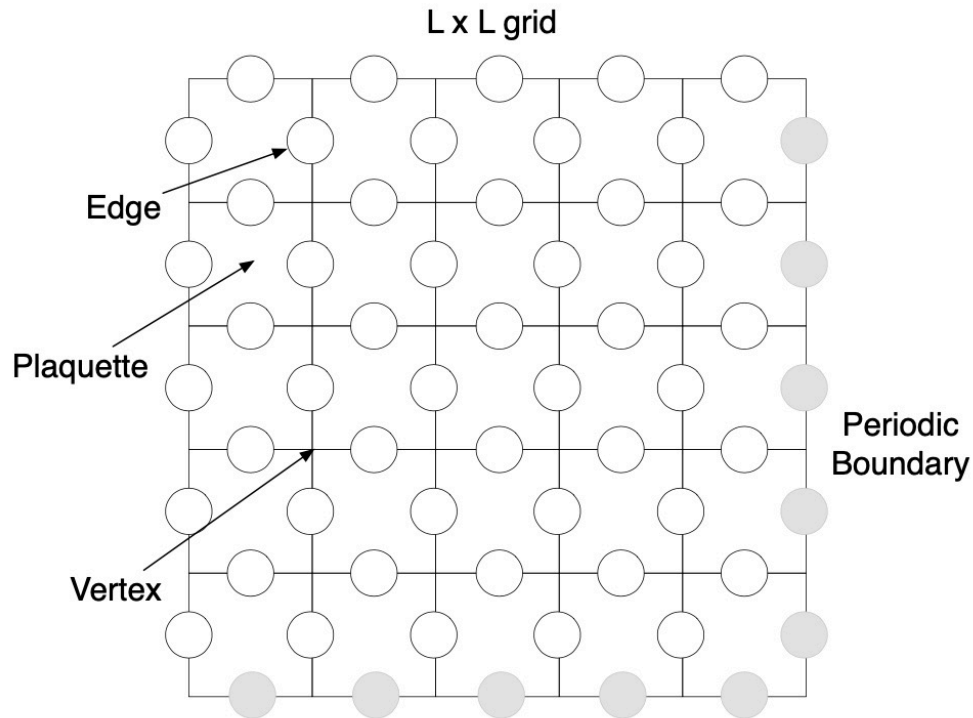
1st example of TQM:

Toric code model
(by Kitaev 1997) $\left(= \mathbb{Z}_2 \text{ gauge theory on torus} \right)$

Hilbert space on torus

$L \times L$ square lattice, $\mathbb{C}^2$ (spin 1/2) on each link

$$\mathcal{H} = \bigotimes_{\text{link } j} \mathbb{C}^2$$



$\mathcal{H}$: Hilbert space

Hamiltonian $H: \mathcal{H} \rightarrow \mathcal{H}$

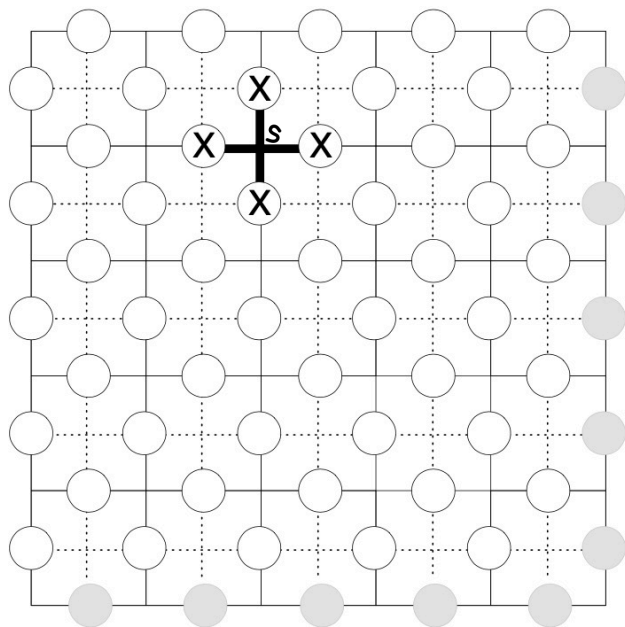
eigenvalues of H are energies E_n of the system

Ground state is the eigenvector with the lowest energy E_0 .

Claim $\left\{ \begin{array}{l} \bullet \text{ The ground state of TC represents } H_1(T^2, \mathbb{Z}_2) = \mathbb{Z}_2 \times \mathbb{Z}_2 \\ \bullet \text{ robust ground states} \\ \bullet \text{ Topological quantum information} \end{array} \right.$

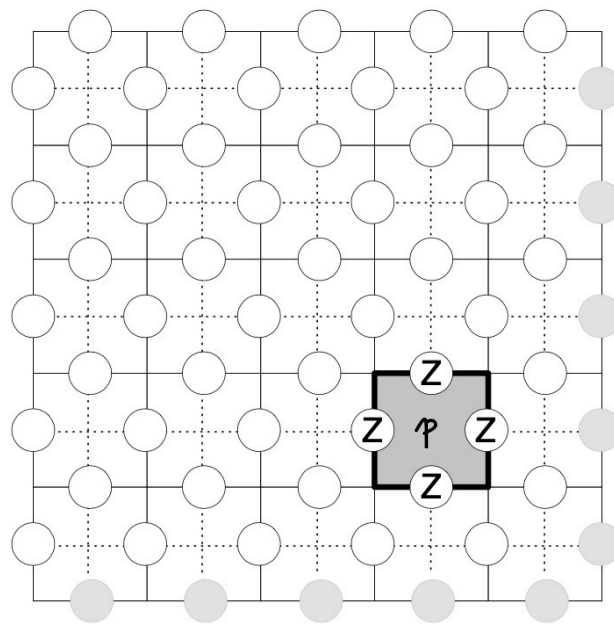
Hamiltonian

$$H = - \sum_{\text{site } s} A_s - \sum_{\text{plaquette } p} B_p$$



$$A_s = \prod_{j \in \text{star}(s)} \sigma_j^x$$

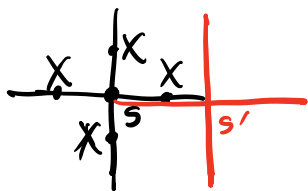
Pauli operator
 $\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$



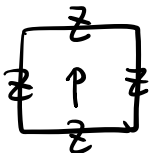
$$B_p = \prod_{j \in \partial p} \sigma_j^z$$

$\sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Algebraic relations of A_5/B_p .

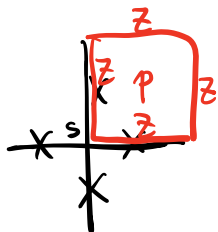


$$\begin{cases} A_s^\dagger = A_s \\ (A_s)^2 = 1 \\ [A_s, A_{s'}] = A_s A_{s'} - A_{s'} A_s = 0 \end{cases} \quad \leadsto A_s \text{ has eigenvalues } \pm 1$$



$$\begin{cases} B_p^\dagger = B_p \\ (B_p)^2 = 1 \\ [B_p, B_{p'}] = 0 \end{cases} \rightsquigarrow B_p = \pm 1.$$

$$[A_s, B_p] = A_s B_p - B_p A_s = 0.$$



$$XZ = -ZX \Rightarrow A_s B_p = B_p A_s$$

$$H = - \sum_s A_s - \sum_p B_p$$

Ground state $|g\rangle$ of TC :

$$A_s |\psi\rangle = B_p |\psi\rangle = +1 |\psi\rangle, \quad \forall s, p.$$

String representation.

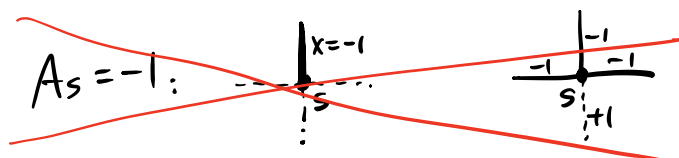
$$A_3 = \pi \sigma^x.$$

$$\sigma^x: \pm 1$$

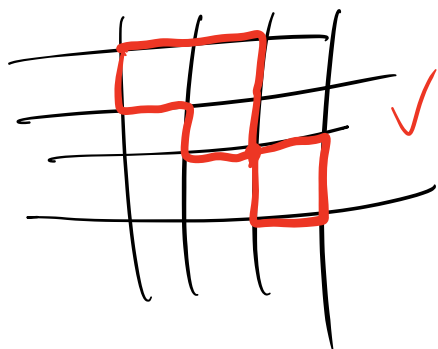
def: $\begin{cases} \sigma_j^x = -1 : \text{ string } \text{---} \\ \sigma_j^x = +1 : \text{ no string } \text{---} \end{cases}$

low energy: $A_5 = 1$:

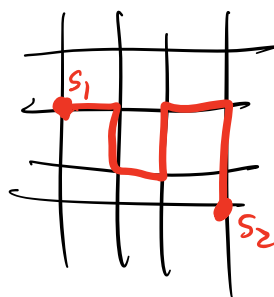
high energy:



ground state : $A_s |\psi\rangle = |\psi\rangle \iff$ closed strings (loops)

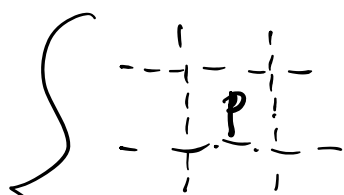


$A_s = +1$
($\forall s$)

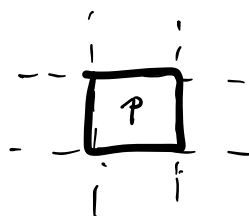


$A_{s_1} = A_{s_2} = -1$
(high energy)

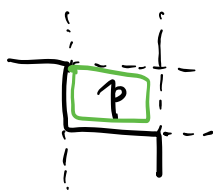
$B_p = \pi \mathbb{Z}$: flip $\sigma^x = +1 \leftrightarrow \sigma^x = -1$



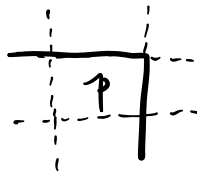
$\xleftrightarrow{B_p}$



create
and/or
small loops.



$\xleftrightarrow{B_p}$



change shapes
of closed loops.

$B_p |c\rangle = |c + \partial p\rangle \pmod{2}$
 $\downarrow$
 $\forall$ closed loop configuration

$$H = - \sum A_s - \sum B_p$$

$\downarrow$ closed loop. $\downarrow$ change shapes.

Assume $|\psi\rangle = \sum_c a_c |c\rangle$, $a_c \in \mathbb{C}$ is the ground state

$$\begin{aligned} \Rightarrow B_p |\psi\rangle &= \sum_c a_c |c + \partial p\rangle = \sum_c a_{c + \partial p} |c\rangle \\ &= |\psi\rangle = \sum_c a_c |c\rangle \end{aligned}$$

$$\Rightarrow a_c = a_{c + \partial p} \text{ for } \forall p$$

The ground state coefficients of c and $c + \partial p$ should be the same.

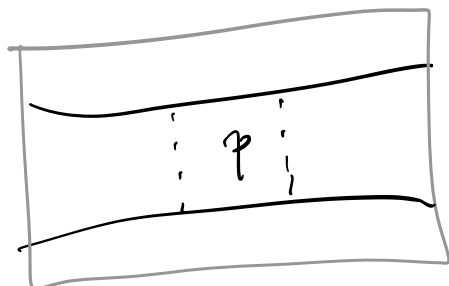
Summary $\begin{cases} A_s = +1 : \text{ conf. } |c\rangle \text{ such that there are only closed loops in } c. \\ B_p = +1 : |c\rangle \text{ and } |c + \partial p \pmod{2}\rangle \text{ have the same coefficient.} \end{cases}$

Ground state subspace of TC
 $= \{ \text{closed loops} \} / \{ \text{boundaries of 2D plaquettes} \}$
 $= H_1(T^2, \mathbb{Z}_2) = \mathbb{Z}_2 \times \mathbb{Z}_2$

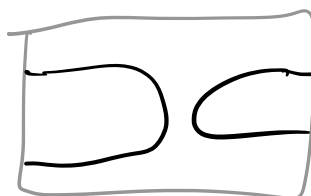
$$|\psi\rangle = | \rangle + | \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \rangle + | \bigcirc \rangle + | \text{---} \rangle + \dots$$

4-fold degeneracy of TC on torus :

$$\begin{cases} |\psi_{00}\rangle = | \rangle + | \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \rangle + | \bigcirc \rangle + \dots \\ |\psi_{10}\rangle = | \text{---} \rangle + | \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \rangle + | \bigcirc \rangle + \dots \\ |\psi_{01}\rangle = | \text{ } \rangle + | \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \rangle + | \bigcirc \rangle + \dots \\ |\psi_{11}\rangle = | \text{ } \rangle + | \text{ } \rangle + | \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix} \rangle + \dots \end{cases}$$



$\rightsquigarrow$



$\rightsquigarrow$



$\rightsquigarrow$



knot/link in (2+1)D spacetime

anyon worldlines in (2+1)D $\longleftrightarrow$ knot/link in 3D

